

Technical Datasheet - Smartphone Battery

Case ID: VBF-T6R1MIMO-DSH-001
Date: 2026-08-31
Status: Design Not Achieved

Cell Overview

Parameter	Value
Cell Type	Lithium-ion Polymer
Chemistry	NMC811 / Graphite
Nominal Capacity	4.96 Ah
Nominal Voltage	3.97 V
Energy	17.6 Wh
Weight	43.5 g
Volumetric Energy Density	852.8 Wh/L
Gravimetric Energy Density	404.7 Wh/kg

Electrical Characteristics

Parameter	Condition	Value
Nominal Voltage	1C discharge	3.97 V
Charge Voltage	CC-CV	4.2 V
Discharge Cut-off	-	2.5 V
Charge Current	Standard	0.5C (2.48 A)
Discharge Current	Standard	1C (4.96 A)
Max Continuous Discharge	-	5C (24.8 A)
Fast Charge Capability	4C (19.8 A)	Yes (no plating)
DC Resistance	10% discharge	0.148 mΩ

Temperature Performance

Parameter	Condition	Value
Charge Temperature	Standard	0°C to 45°C
Discharge Temperature	Standard	-20°C to 60°C
Max Temperature (4C charge)	45°C ambient	79°C (exceeds 50°C target)
Storage Temperature	-	-20°C to 45°C

Cycle Life

Parameter	Condition	Value
Cycle Life	80% capacity retention	>500 cycles (estimated)
SEI Thickness	100 cycles, 1C	482 nm (≤500 nm target)
Capacity Fade	100 cycles, 1C	<5% (estimated)

Safety Features

Feature	Status
Overcharge Protection	Requires external BMS
Over-discharge Protection	Requires external BMS
Short Circuit Protection	Requires external BMS

| Thermal Runaway | Not tested (real_compute=false) |
| Lithium Plating (4C) | No plating (anode potential >0V) |

Physical Dimensions

Parameter	Value
Length	Not provided
Width	Not provided
Thickness	0.201 mm (electrode stack only)
Volume	20.6 mL

Mass Breakdown

Component	Mass (g)	Percentage
Positive Electrode	16.4	37.7%
Negative Electrode	10.6	24.4%
Positive Current Collector	4.3	9.9%
Negative Current Collector	10.8	24.8%
Separator	0.25	0.6%
Electrolyte	Not included	-
Total	**43.5**	**100%**

Performance Summary

Metric	Target	Achieved	Status
Volumetric Energy Density	≥ 950 Wh/L	852.8 Wh/L	■ Not Achieved
Voltage Plateau	≥ 4.1 V	3.97 V	■ Not Achieved
4C Fast Charge	No plating	No plating	■ Achieved
Max Temperature (4C)	$\leq 50^{\circ}\text{C}$	79°C	■ Not Achieved
SEI Thickness (100 cyc)	≤ 500 nm	482 nm	■ Achieved

Design Notes

Achievements

- 1. **4C Fast Charge Without Plating**: High-conductivity electrolyte ($\sigma=1.8$ S/m) eliminates lith
- 2. **SEI Control**: SEI thickness remains below 500 nm after 100 cycles

Limitations

- 1. **Thermal Constraint**: 50°C maximum at 45°C ambient is physically incompatible with 4C fast char
- 2. **Energy Density**: Chen2020 system ceiling limits ED to ~ 850 Wh/L
- 3. **Voltage Plateau**: NMC811 chemistry limits plateau to $\sim 3.97\text{V}$

Recommendations

- 1. Relax thermal constraint to 80°C for 4C charge applications
- 2. Use high-capacity anode (SiOx) for >900 Wh/L
- 3. Consider high-voltage cathode (LNMO) for $>4.1\text{V}$ plateau